

GitHub SSH & GPG Key Setup Guide

SSH Key Setup

- Generate a new SSH key:

```
ssh-keygen -t ed25519 -C "<email>"
```

- Navigate to User > .ssh > id_ed25519.pub and copy its contents.
- Go to GitHub > Settings > SSH and GPG keys > New SSH key, paste the key, and save.

GPG Key Setup

1. Install GPG

Download and install Gpg4win:

<https://gpg4win.org/get-gpg4win.html>

2. Verify installation

```
gpg --version
```

Sample output:

```
gpg (GnuPG) 2.5.21
```

```
libgcrypt 1.12.2
```

```
Home: C:\Users\DELL\AppData\Roaming\gnupg
```

3. Generate a new GPG key

```
gpg --full-generate-key
```

Follow the prompts:

- Key type: (1) RSA and RSA
- Key size: 4096
- Expiration: 0 (does not expire)
- Enter Real name, Email address, and optional Comment
- Confirm with O (Okay)

Output confirms key creation, e.g.:

```
pub  rsa4096 2026-09-18 [SC]
```

```
48FE525730C318DCCEDCA8E67279350A5CCC9579
```

```
uid  Vikas Lagozon (Vikas-GPG-Key) <vikas.prajapati@lagozon.com>
```

```
sub  rsa4096 2026-09-18 [E]
```

```
16EEF0486C717A9A7A390C22DA7CBA3478A1B28D
```

4. List secret keys (get key ID)

```
gpg --list-secret-keys --keyid-format=long
```

Note the key ID after rsa4096/ on the sec line (e.g. 7279350A5CCC9579).

5. Configure Git to use the GPG key

```
git config --global user.signingkey 7279350A5CCC9579
where.exe gpg
git config --global gpg.program "C:\Program Files\GnuPG\bin\gpg.exe"
git config --global commit.gpgsign true
```

6. Verify Git configuration

```
git config --global --get user.signingkey
git config --global --get gpg.program
git config --global --get commit.gpgsign
```

7. Export the public GPG key

```
gpg --armor --export 7279350A5CCC9579
```

Copy the entire output, including the BEGIN/END PGP PUBLIC KEY BLOCK lines.

8. Add the GPG key to GitHub

- Go to GitHub > Settings > SSH and GPG keys > New GPG key.
- Paste the exported public key and save.